

Interactive Deformable Display

Freeform shapes with novel liquid metal wirings

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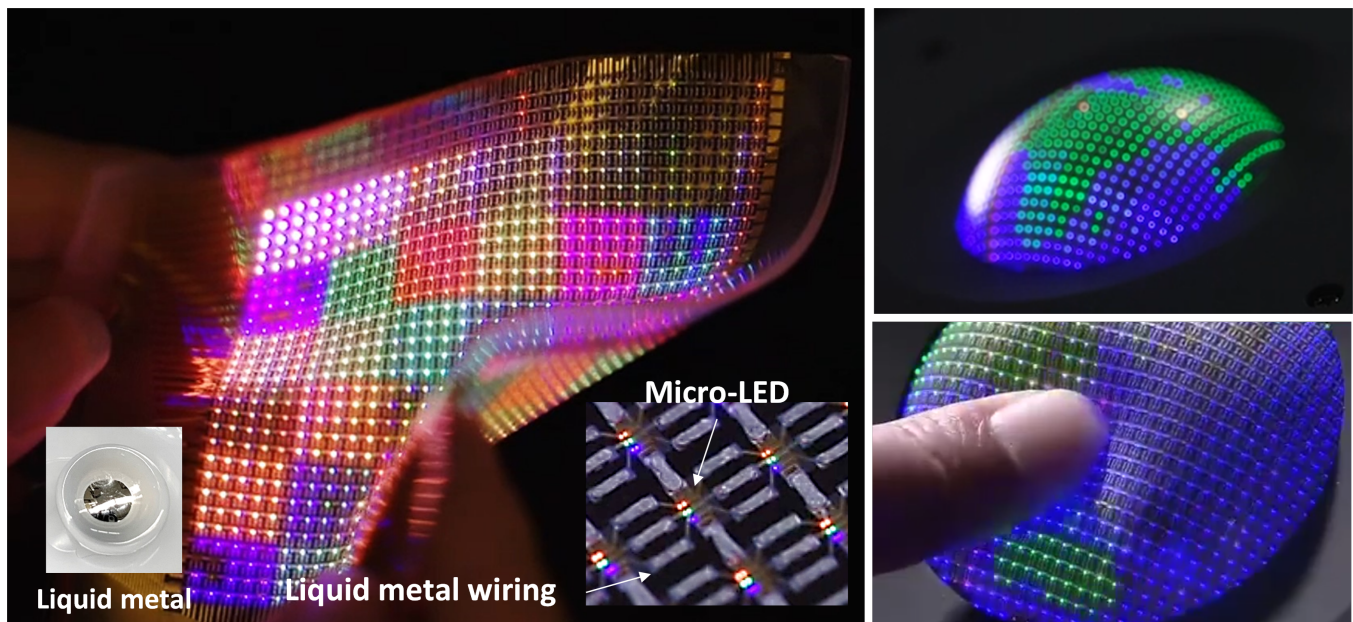


Figure 1: Photograph of the developed highly stretchable LED display using novel liquid metal wiring under deformation conditions, such as spherical, twisted, and rounded shapes.

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Abstract

This study demonstrates a directly controllable interactive deformable display. The high stretchability of the display, achieved through novel liquid metal wiring, enables three-dimensional dynamic deformation, such as concave, convex, and folding shapes. An integrated strain sensor on a stretchable display calculates its display deformation and allows direct interaction between users and the display device. This unprecedented display type provides a completely novel interactive experience with dynamic deformation.

CCS Concepts

• Hardware → Displays and imagers.

Keywords

Deformable display, direct interaction, three-dimensional (3D) shapes

ACM Reference Format:

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1 INTRODUCTION

Displays are essential devices for human–media interactions. Stretchable displays have been studied extensively in recent years. This stretchable technology is considered a new generation of electronics that drastically changes the style from the current rigid, hard, and flat shapes to an organic style, which can promote a pleasing design [Hong et al. 2017; Koo et al. 2018; Rogers et al. 2010]. The non-flat three-dimensional (3D) curved shapes result in warm tones and create a cozy welcoming atmosphere. Realizing stretchable displays allows for pleasant organic design with fully free shapes and integrability onto 3D shapes for an immersive and realism-based content experience. Stretchable displays have been thoroughly studied; however, they remain in the early stages of development and require technological advancements. The realization of highly stretchable and directly controllable full dome or sphere-shaped displays requires a high stretchability of 50% device strain.

We present a new type of directly controllable interactive stretchable display with dynamic deformation and direct interactions that enables a new experience of non-flat 3D curve shapes. The developed stretchable LED display realized consistent and stable stretchable electrical connections by adopting novel liquid metal wiring. We confirmed that the display exhibits excellent dynamic deformation with a high stretchability of up to 50%. The display is mechanically stable when applied to 3D curved shapes, such as domes, or when folded in half like a handkerchief, without any disconnection of lines or pixels. In addition, the integrated sensor enables direct control to enrich the visual experience. The developed display is a major step in expanding innovative soft electronics and creating new user experiences.

2 OVERVIEW OF SYSTEM

3D deformable stretchable LED display using liquid metal wiring was developed to realize high stretchability of display up to 50% device strain. Figure 1 depicts photographs of the developed deformable LED display (32×32 RGB micro-LED display with dimensions of 64×64 mm) with deformed shapes such as folding, dome, and pressing by finger.

Liquid metal materials (LMs) are liquid-phase metals at room temperature. LMs have emerged as promising candidates for fabricating highly stretchable and conductive conductors. However, handling LMs can be challenging owing to their reactivity and

non-wettability, particularly in the fine patterning of display pixel interconnections. Their reactivity with other metals is another issue. We successfully developed an LM fine-patterning technique for LM materials by modifying its properties to achieve viscosity and good wettability. In addition, we developed an interfacial metal composition between the LMs and contact metals to establish reliable electrical contacts and ensure long-term reliability [Miyakawa et al. 2025]. The LM used in this study is a Ga-In-based alloy. Because reliable display deformation requires durability, we optimized the display structure using a strong adhesive layer to prevent delamination and displacement issues [Miyakawa et al. 2022].

The LEDs were mounted directly onto isolated film pixels formed on a stretchable substrate. Stretchable LM wiring for LEDs was connected using metal electrodes on an island-shaped flexible polyimide film. The developed LM-based display demonstrated a stretchability of approximately 50%. Furthermore, we validated the reliability of the display by subjecting it to repetitive deformation over 12,000 cycles and confirming its long-term storage of more than 300 days. In addition, the LM wirings possessed self-repairing capabilities in the event of excessive stretching because the LM materials maintained a liquid state.

Direct controllable function was realized by sensing resistance changes during display deformation as shown in Figure 2. An integrated resistance sensor can detect the deformation of a display, thereby enabling direct interaction. This direct interaction enables significant soft display interaction with an unprecedented interface.

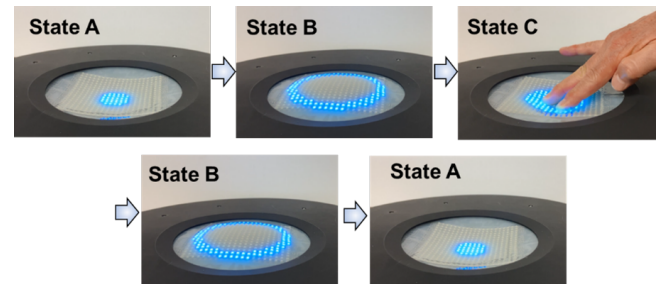


Figure 2: Demonstration of an interactive deformable display showing a controllable display by pressing. The display image changes from states A, B, and C, and the pressing action triggers a different image showing states C, B, and A.

3 DEMO EXPERIENCE

At SIGGRAPH 2025 Emerging Technologies, we will showcase the proposed interactive deformable display with high stretchability enabled by liquid metals. Attendees interested in interactive demonstrations will experience a new concept of directly controllable, interactive freeform deformable displays. Our deformable display provides dynamic deformation and unique content because the display is deformed using a customized inflating or deflating air apparatus. Furthermore, the integrated sensors calculate the deformed shapes and display controlled images. Therefore, the proposed device provides a novel visual experience.

4 CONCLUSION

We developed a directly controllable interactive stretchable display with dynamic deformation and direct interactions to provide a new experience with non-flat 3D curved shapes. Stretchable displays represent cutting-edge technology with the potential to revolutionize visual experiences and ambient information display applications.

Acknowledgments

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